Fretboard Modeller

User guide – design a fretboard, machine it, and cut the fret slots

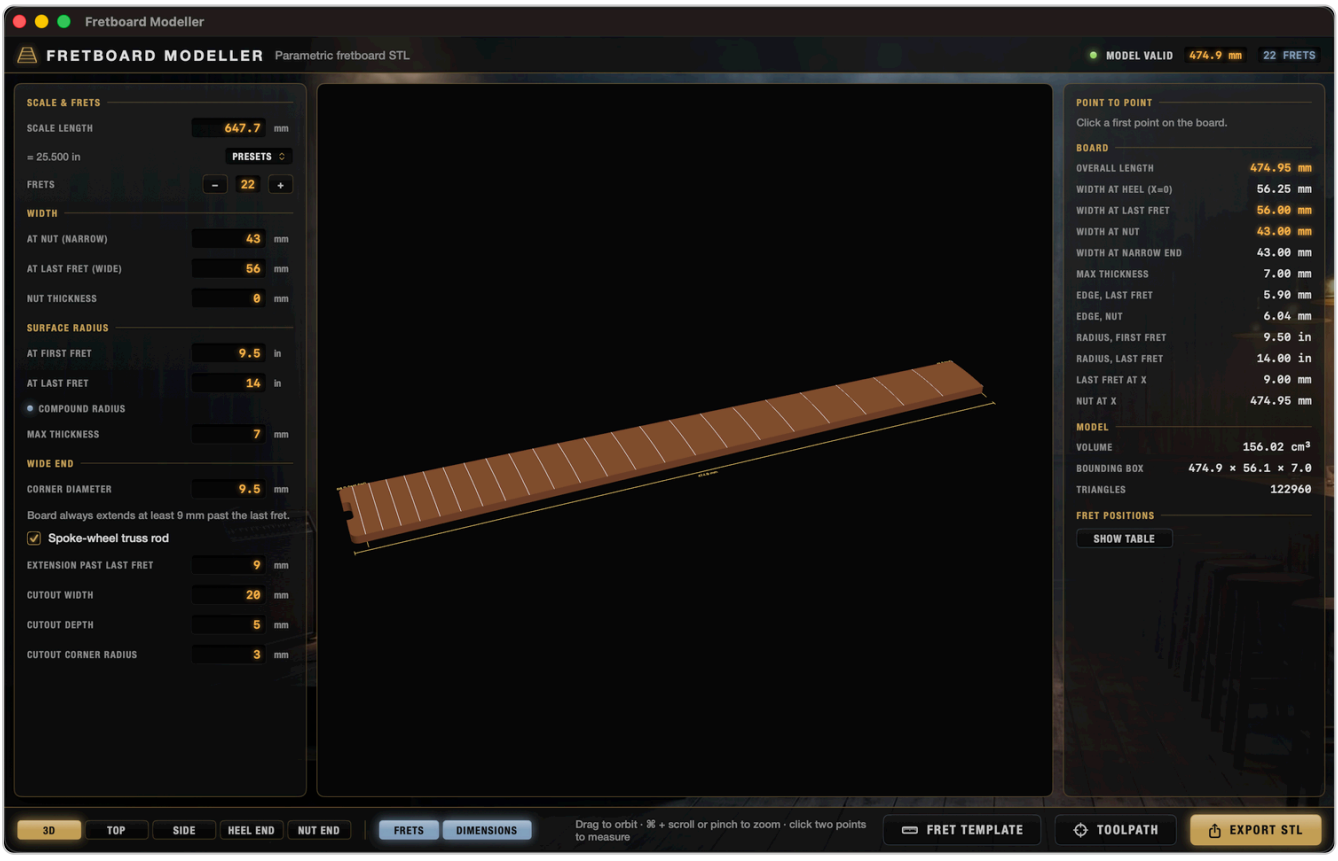
1. Overview

Fretboard Modeller has three tools that share one set of board dimensions:

Tool	Opened from	Produces
Designer	Main window	3D fretboard, measurements, binary `.stl` (mm)
Toolpath	TOOLPATH button	WorkBee Z1+ G-code (`.nc`) with animated playback
Fret Template	FRET TEMPLATE button	Marker-hole G-code for the chop-saw fret slot jig

Launch it from Spotlight (⌘ Space, “Fretboard Modeller”) after running `./make_app.sh`. Both secondary windows have a ← **DESIGNER** button that closes them and returns you to the main window.

2. Designer



Inputs

Field	Meaning
Scale length	In mm; the inch equivalent is shown, and PRESETS offers common scales.
Frets	Number of frets (1–36).
At nut / at last fret	Board width at the nut and at the last fret; the board tapers linearly between them.
Nut thickness	Extra length added past the nut at the narrow end (default 0 mm).
Surface radius	Radius in inches at the first and last fret. Equal values = constant radius; different values = compound radius.
Max thickness	Thickness on the centre line (7 mm default). With a compound radius the centre-line thickness stays constant and the edges get thinner toward the wide end.
Corner diameter	Rounding of the two wide-end corners (9.5 mm default). The narrow end has hard corners.
Extension past last fret	The board always extends at least 9 mm beyond the last fret.
Spoke-wheel truss rod	On by default. Adds a cutout centred on the heel edge: width 20 mm, depth 5 mm, inside corner radius 3 mm (all editable).

Orientation

The widest end (heel) sits at X = 0; the narrowest end points toward +X. Y is across the board, centred on 0. Z is up, with the bottom of the board at Z = 0.

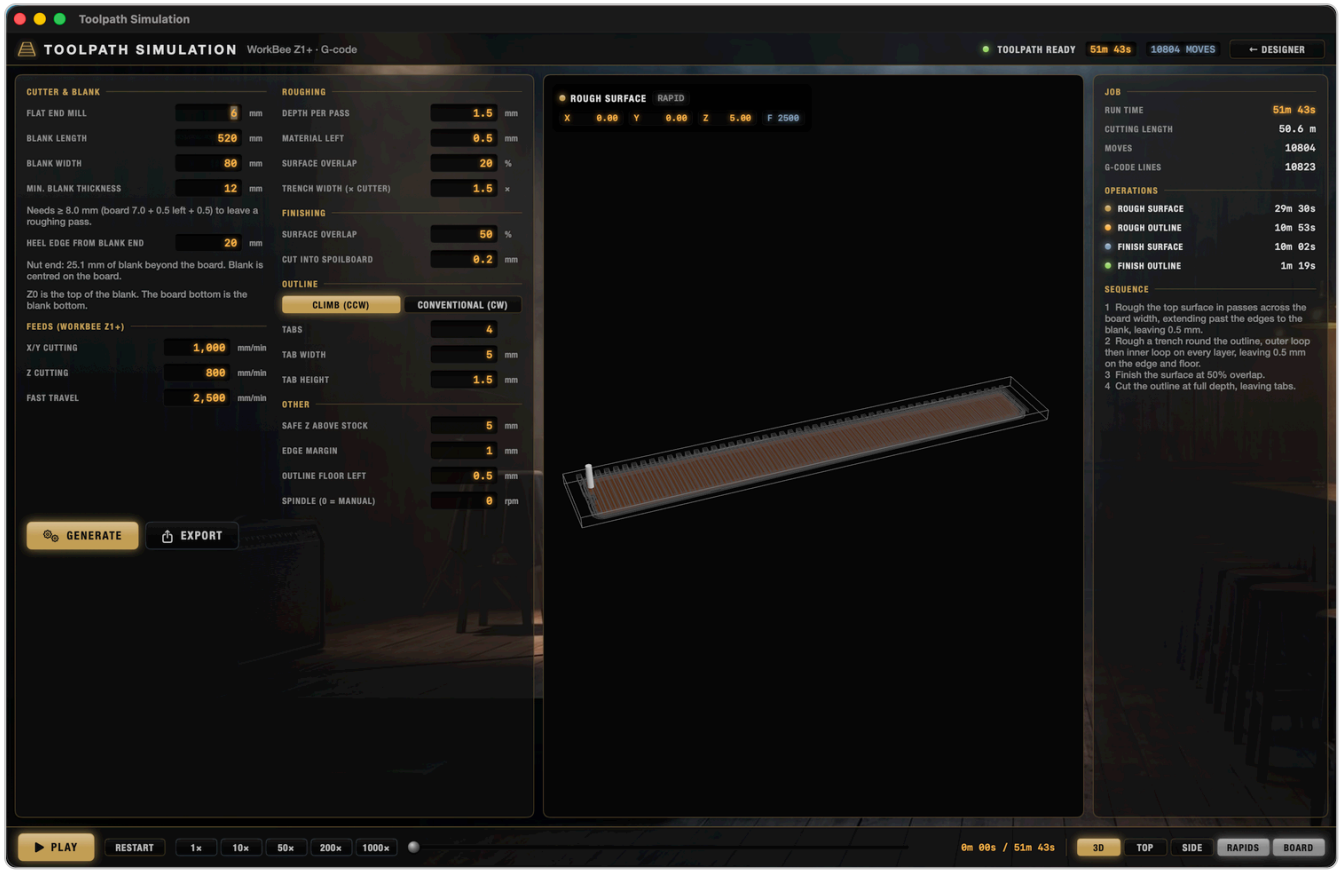
Viewer and measurements

- Drag to orbit; ⌘ + scroll or pinch (Magic Mouse / trackpad) to zoom.
- Buttons along the bottom set 3D, TOP, SIDE, HEEL END and NUT END views and toggle fret lines and dimension lines.
- Click two points on the board for a point-to-point distance.
- The right panel shows overall length, widths, edge thicknesses, radii, volume, bounding box and triangle count; SHOW TABLE lists every fret position.

Exporting

EXPORT STL writes a binary STL in millimetres. The mesh is checked to be watertight before export is allowed; if a setting is invalid the status badge turns red and tells you why.

3. Toolpath (CNC)



Press **TOOLPATH** to open the toolpath window. Set the blank and options on the left, press **GENERATE**, review the run in the viewer, then **EXPORT** to save the G-code. The status badge shows *OUT OF DATE* if you change the board after generating – generate again before exporting.

Coordinate system

X0 = heel edge of the board, Y0 = board centre line, **Z0 = top of the blank**. The bottom of the board is the bottom of the blank. Cutter: 6 mm flat end mill. Set these work offsets on the machine before running.

Blank

Setting	Notes
Blank length / width	Defaults 520 × 80 mm. The blank is centred on the board's centre line.
Min. blank thickness	Default 12 mm. Must be at least board thickness + material left + 0.5 mm (8.0 mm for the default 7 mm board) so there is at least one roughing pass. Thickness blanks beforehand to about this value.
Heel edge from blank end	Default 20 mm; the nut-end margin is calculated and shown.

> The path is clipped to the blank and the app rejects a blank that cannot contain the board (1 mm all round). An amber warning appears if the blank is too tight for the outline trench.

Sequence

1. **Rough surface** – raster passes across the board width, running past the board edges, 1.5 mm deep, leaving 0.5 mm, 20 % overlap.
2. **Rough outline** – a trench round the board 1.5× the cutter width wide, cut as two loops on every depth layer: the outer loop first and the loop beside the board last, so the board edge gets the lightest cut. Leaves 0.5 mm on the edge and floor.
3. **Finish surface** – 50 % overlap, removing the last 0.5 mm.
4. **Finish outline** – full-depth cut round the board, leaving tabs.

The order (surfaces and trench first, final outline last) is chosen to reduce splitting.

Tabs and outline direction

- **Tabs** hold the board in the blank on the final cut: default 4, each 5 mm wide and 1.5 mm tall. Cut them away by hand afterwards. Set tabs to 0 if you hold the blank down some other way (tape and superglue work as double-sided tape).
- **Climb (CCW)** or **Conventional (CW)** sets the outline direction. If you see splits on one side of the board, switch direction.

Feeds

X/Y cutting 1000 mm/min, Z cutting 800 mm/min. **Fast travel defaults to 2500 mm/min as a placeholder – set it to your machine's configured maximum.** Spindle RPM 0 means manual spindle control (no M3/M5 written).

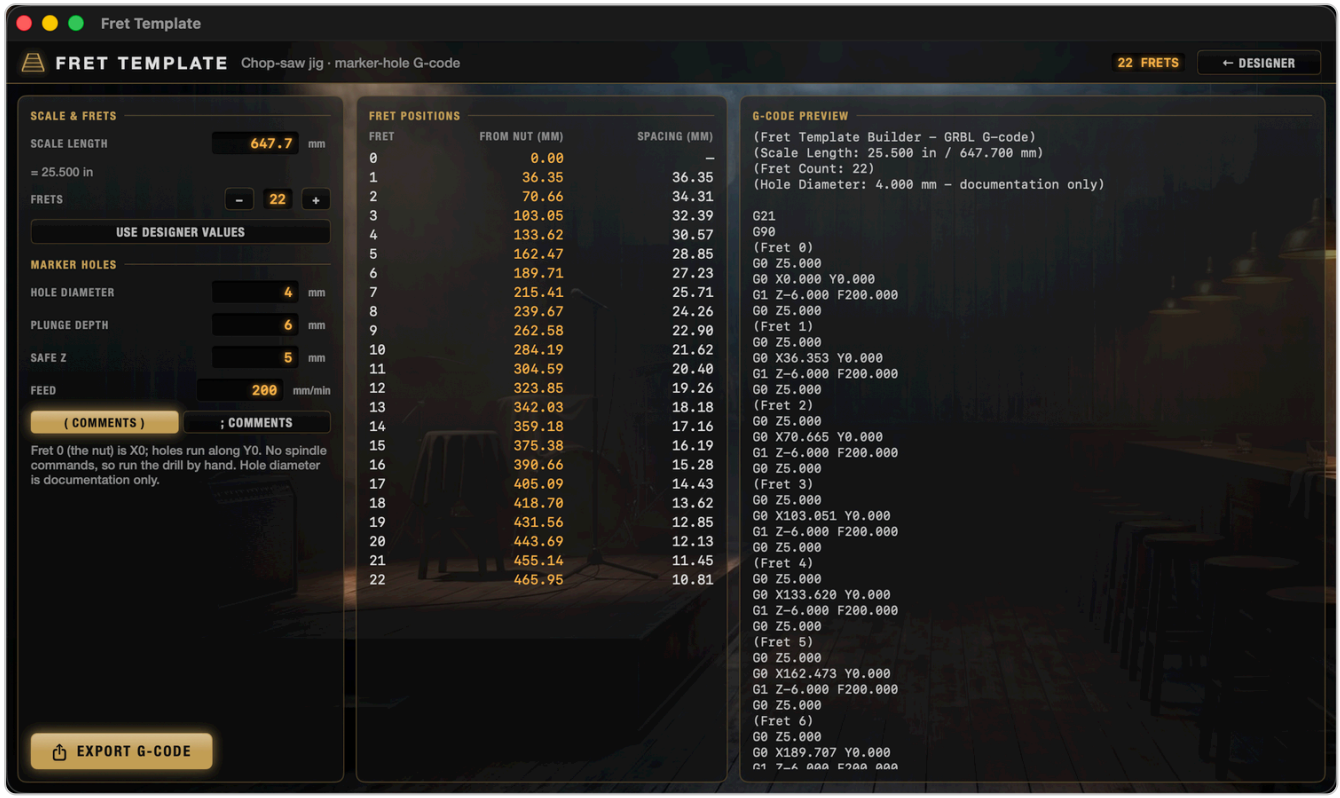
Playback

Play the run at 1×, 10×, 50×, 200× or 1000× with the slider; the cutter and live X/Y/Z/feed readout follow the path. Views: 3D / TOP / SIDE; toggle RAPIDS and BOARD. Path colours: gold = rough surface, amber = rough outline, blue = finish surface, green = finish outline. Playback shows the cutter path, not material removal.

G-code format

Duet / RepRapFirmware: `G21`, `G90`, `G17`, explicit `G0` / `G1` with feed rates, header comments describing the job, and a section comment per operation. Check the output against your machine's configuration before the first run, and test above the stock.

4. Fret Template



Creates the drilled-hole template used to position a jig for cutting fret slots on a chop saw. Fret positions come from $d = L \cdot (1 - 1/2^{n/12})$, measured from the nut. Fret 0 is X0 and all holes lie on Y0.

Setting	Default
Scale length / frets	Taken from the designer; **USE DESIGNER VALUES** resets them
Hole diameter	4 mm (written as a comment only – drill with a 4 mm bit)
Plunge depth	6 mm
Safe Z	5 mm
Feed	200 mm/min
Comments	( ) or ; style

The table lists each fret's distance from the nut and spacing from the previous fret; the preview shows the G-code. **EXPORT G-CODE** saves `fret-slot-template-<scale in>-<frets>.nc`. The output is GRBL-style (ends with `M30`) and has no spindle commands, so run the drill by hand. Holes are at the fret centrelines; no blade-kerf offset is applied.

5. Building and testing

`./make_app.sh`	Builds the release app and installs it to ~/Applications (Spotlight-launchable).
`swift run FretboardApp`	Runs from the command line.
`swift test`	Runs the test suite (see the README for the Command-Line-Tools-only flags).

6. Troubleshooting

Symptom	Fix
Status badge red in the designer	Read the message; e.g. heel narrower than nut, radius too small for the width, or cutout too big.
Toolpath says blank is too small / too thin	Increase the blank size or thickness; see the minimum shown under Min. blank thickness.
Toolpath “out of date”	Press GENERATE after changing the board or settings.
Splitting on one edge	Switch the outline direction (climb / conventional).
Travel feels slow or too fast	Set Fast travel to your machine's real maximum.